```java
import java.util.Scanner;

public class QuickSort {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Get array size from user
        System.out.print("Enter number of elements: ");
        int n = scanner.nextInt();

        // Create and initialize an array
        int[] arr = new int[n];
        System.out.println("\nEnter element values:");
        for (int i = 0; i < n; i++) {
            arr[i] = scanner.nextInt();
        }

        // Sort the array using Quick Sort
        quickSort(arr, 0, n - 1);

        // Print the sorted array
        System.out.println("\nSorted array:");
        for (int i : arr) {
            System.out.print(i + " ");
        }
        System.out.println();
    }

    private static void quickSort(int[] arr, int low, int high) {
        if (low < high) {
            // Partition the array
            int pivotIndex = partition(arr, low, high);

            // Sort recursively the sub-arrays
```

```java
            quickSort(arr, low, pivotIndex - 1);
            quickSort(arr, pivotIndex + 1, high);
        }
    }

    private static int partition(int[] arr, int low, int high) {
        // Choose the last element as pivot
        int pivot = arr[high];
        int i = (low - 1);

        // Partitioning loop
        for (int j = low; j <= high - 1; j++) {
            if (arr[j] <= pivot) {
                i++;
                int temp = arr[i];
                arr[i] = arr[j];
                arr[j] = temp;
            }
        }

        // Place the pivot element at its correct position
        int temp = arr[i + 1];
        arr[i + 1] = arr[high];
        arr[high] = temp;
        return (i + 1);
    }
}
```